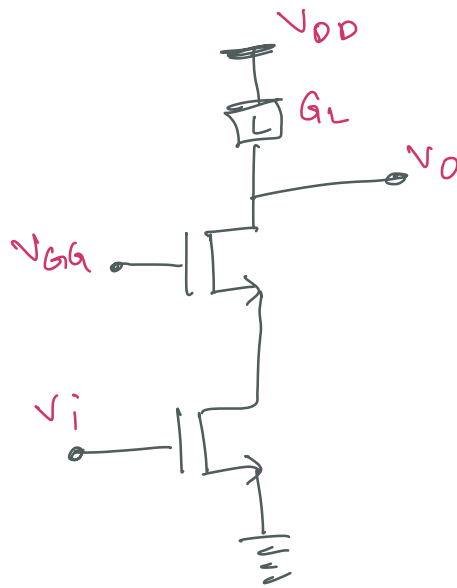


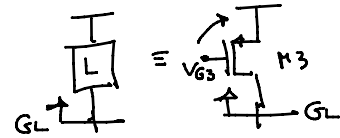
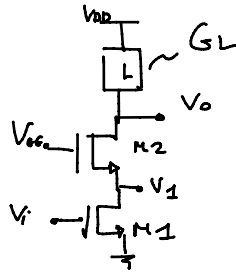
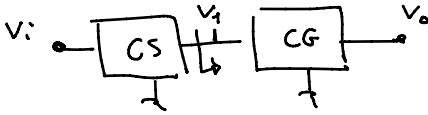
Lecture-9

→ This is the continuation of CASCODE STAGE.

CASCODE, FOLDED
cascode.



Revanth Reddy
Pannala
EBIT, Unibo

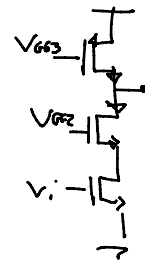


$$A_V = \frac{v_o}{v_i} = \frac{v_o}{v_1} \times \underbrace{\frac{v_1}{v_i}}_{A_{V1}}$$

$$G_{L1} = G_{icf} = \frac{G_{T2} \cdot G_{L2}}{g_{o2} + G_{L2}} = \frac{G_{T2} \cdot g_{o3}}{g_{o2} + g_{o3}}$$

$$A_{V1} = A_{V_{CS}} = -\frac{g_{m1}}{g_{o1} + G_{L1}}$$

$$A_{V1} = -\frac{g_{m1} (g_{o2} + g_{o3})}{g_{o1} (g_{o2} + g_{o3}) + G_{T2} \cdot g_{o3}} \approx -\frac{g_{m1} (g_{o2} + g_{o3})}{G_{T2} \cdot g_{o3}}$$



$$g_{o2} = \lambda_n \cdot I_{D2}$$

$$g_{o3} = \lambda_p \cdot I_{D3}$$

$$I_{D1} = I_{D2} = I_{D3}$$

$$\lambda_n \approx \lambda_p$$

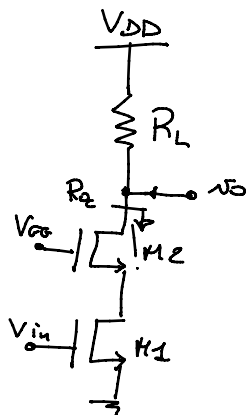
$$A_{V1} \approx -2 \frac{g_{m1}}{g_{m2}} = -2 \frac{\sqrt{2 \beta_1 I_{D1}}}{\sqrt{2 \beta_2 I_{D2}}} = -2 \frac{\sqrt{S_1}}{\sqrt{S_2}}$$

lecture-8

$$A_{V_{CG}} = \frac{v_o}{v_1} = \frac{G_{T2}}{G_L + g_{o2}} = \frac{G_{T2}}{g_{o3} + g_{o2}}$$

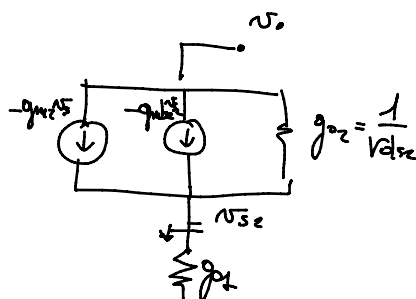
$$A_V = \frac{v_o}{v_i} \cdot \frac{v_1}{v_i} = -\frac{G_{T2}}{g_{o3} + g_{o2}} \cdot \frac{g_{m1} (g_{o3} + g_{o2})}{g_{o3} \cdot G_{T2}} = -\frac{g_{m1}}{g_{o3}}$$

lecture-9



$$G_L \ll g_{o3}, g_{o2}, g_{p1}$$

$$R_{oc} \Big|_{\substack{v_{in}=0 \\ R_L=\infty}} = R_{oc} \Big|_{\substack{v_{in}=0 \\ R_L=\infty}} = R_{oc} \Big|_{\substack{v_{G1}=0 \\ R_L=\infty}}$$

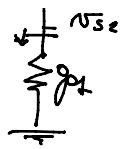


$$R_D = r_{ds2} \{ 1 + G_{T2} \cdot r_{ds1} \}$$

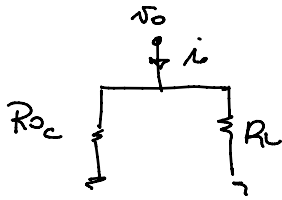
$$G_{T2} \approx g_{m2} \quad \left. \begin{matrix} g_{m2} \gg 1 \end{matrix} \right\} \Rightarrow R_{oc} \approx g_{m2} \cdot r_{ds2} \cdot r_{ds1}$$

$$g_m \gg g_o$$

→



$$\frac{g_{m2}}{g_{o1}} \gg 1 \Rightarrow R_{oc} \approx g_{m2} \cdot v_{ds2} \cdot r_{ds1}$$



$$G_{||} = \frac{1}{R_{||}} = \frac{1}{R_{oc}} + G_L = \frac{g_{o1} \cdot g_{o2}}{g_{m2}} + G_L$$

$$\left[G_{||} = \frac{g_{o1} \cdot g_{o2} + G_L \cdot g_{m2}}{g_{m2}} \right]$$

$$A_{V_{casc} R_L} = A_{V_{1CS}} \cdot A_{V_{2CG}}$$

$$A_{V_{1CS}} = -\frac{g_{m1}}{g_{o1} + G_{L1}} ; G_{L1} = G_{icg} = \frac{G_{T2} \cdot G_L}{g_{o2} + G_L} ; A_{V_{2CG}} = \frac{G_{T2}}{G_L + g_{o2}}$$

$$A_{V_{casc}} = -\frac{g_{m1}}{g_{o1} + \frac{G_{T2} \cdot G_L}{g_{o2} + G_L}} \cdot \frac{G_{T2}}{G_L + g_{o2}} = -\frac{g_{m1} \cdot G_{T2}}{g_{o1}(g_{o2} + G_L) + G_{T2} \cdot G_L}$$

approx. $G_{T2} \sim g_{m2}$

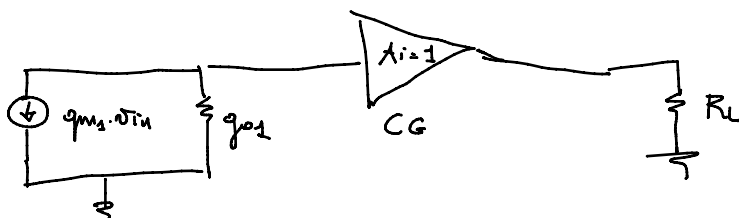
$$A_{V_{casc}} \Big|_{R_L = r_{ds3}} = -\frac{g_{m1}}{g_{o3}}$$

$$A_{casc} \Big|_{R_L} \approx -\frac{g_{m1} \cdot g_{m2}}{g_{o1} \cdot g_{o2} + g_{m2} \cdot G_L} \sim \frac{1}{G_{||}}$$

$$A_{casc} \Big|_{R_L} = -g_{m1} \cdot R_{||}$$

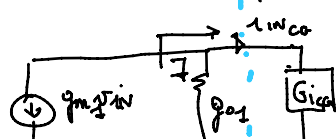
CS.

v_{in}

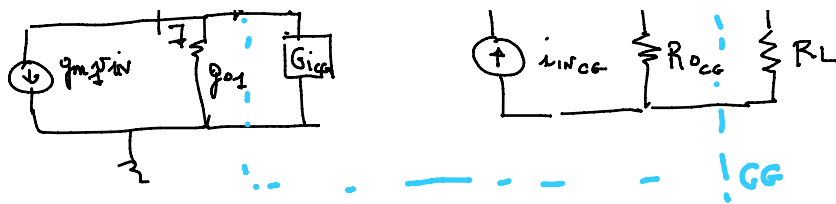


$$R_{oc} \sim \frac{g_m}{g_o} \sim 100$$

v_{in}



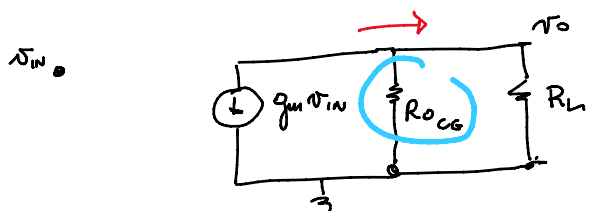
$$G_L \ll g_o$$



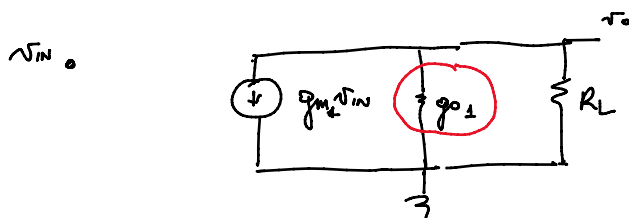
$$G_L \ll g_o$$

$$R_L \gg r_{ds}$$

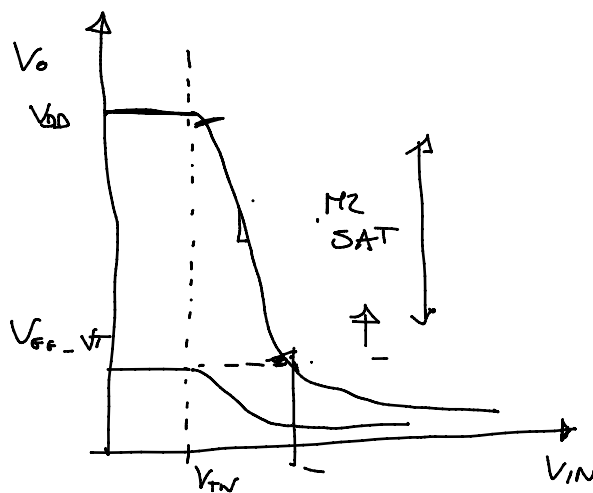
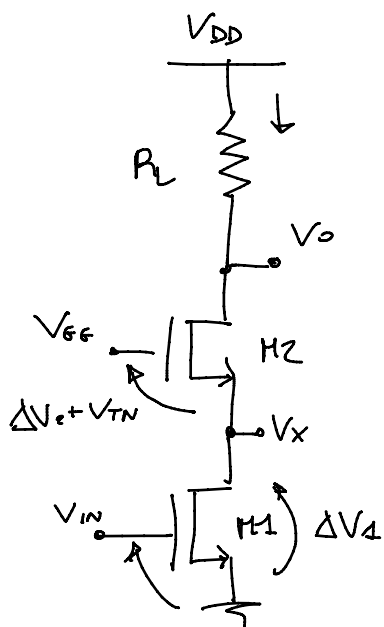
$$G_{iCG} \gg g_{o1} ; i_{inCG} \approx -g_{m1} \cdot v_{in} ; v_o = -g_{m1} \cdot v_{in} \cdot R_{||}$$



CASCODE
 $\frac{g_m}{g_o} \uparrow$ SAT



C.S.



$$M1 \text{ SAT} \Rightarrow V_x \geq V_{in} - V_{TN}$$

$$M2 \text{ SAT} \Rightarrow V_o - V_x \geq \underbrace{V_{GG} - V_x - V_{TN}}_{\Delta V_2} \Rightarrow V_o \geq \underline{V_{GG} - V_{TN}}$$

$$[V_{GG_LIN} = \Delta V_1 + \Delta V_2 + V_{T2}]$$

$$\Delta V_2 = \sqrt{\frac{2I_{D1}}{\beta_1}} \approx [0.1 \div 0.4]$$

$$V_o \geq \Delta V_1 + \Delta V_2 + V_T - V_T$$

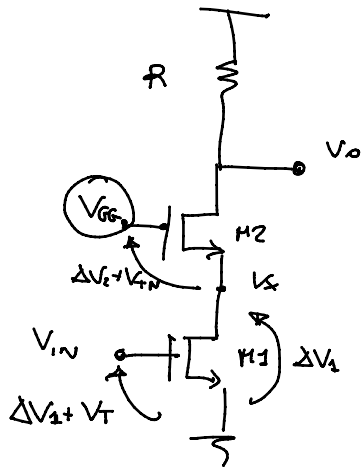
$$V_{o_MIN} = \Delta V_1 + \Delta V_2$$

$$V_{O_{MIN}} = \sqrt{2I_D} \left[\frac{1}{\sqrt{\mu_1}} + \frac{1}{\sqrt{\mu_2}} \right]$$

$$A_v = -50$$

$$[v_i = \pm 20mV]$$

$$v_o = \pm 1V$$



M2 SAT

$$V_o \geq V_{GG} - V_{T2} \triangleq V_{O_{MIN}} \downarrow \Rightarrow V_{GG} \downarrow$$

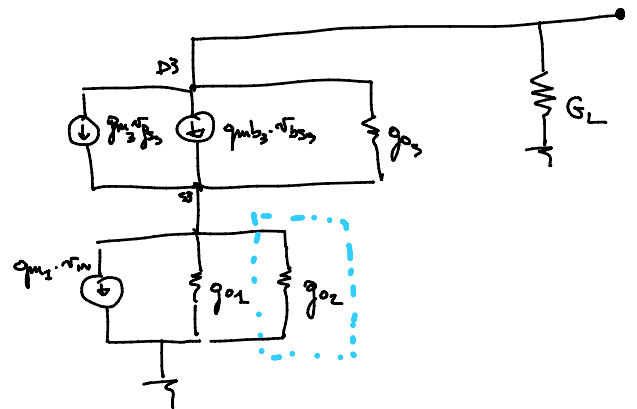
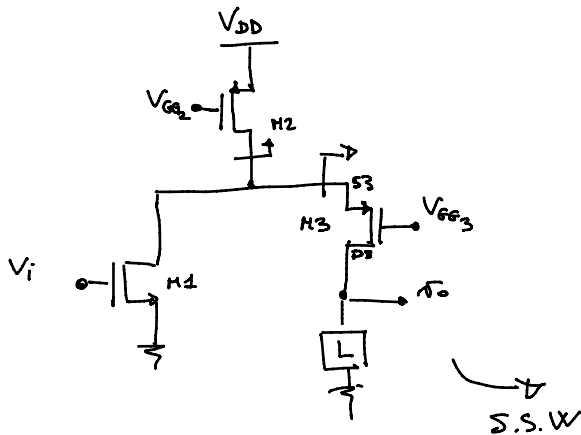
M1

$$V_{GG} \geq \Delta V_2 + V_T + \Delta V_1$$

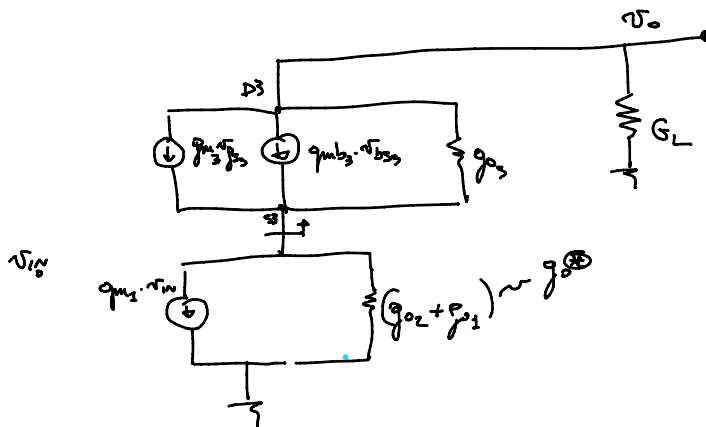
$$V_{GG} \geq \Delta V_2 + V_{T2} + V_{iN} - V_{T1}$$

$$V_{iN} \leq \underbrace{V_{GG} - \Delta V_2 + V_{T2} - V_{T1}}_{V_{i_{MAX}}} \triangleq V_{i_{MAX}}$$

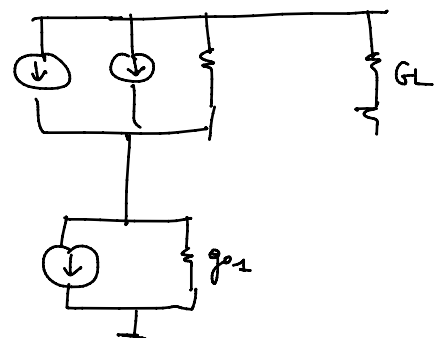
$$V_{i_{MAX}} \uparrow \quad V_{GG} \uparrow$$



SSC → Folded Cascode



SSC Cascode



$$A_{V_{CS}} = \frac{-g_{m1}}{g_{o1} + G_L}$$

$$A_{V_{FC}} = \frac{-g_{m1}}{g_{o1} + g_{o2} + G_{icg}}$$

$$\underline{A_{V_{CASC}}} = -g_{m1} \left(\underline{R_{o_{CG}}} \parallel R_L \right)$$

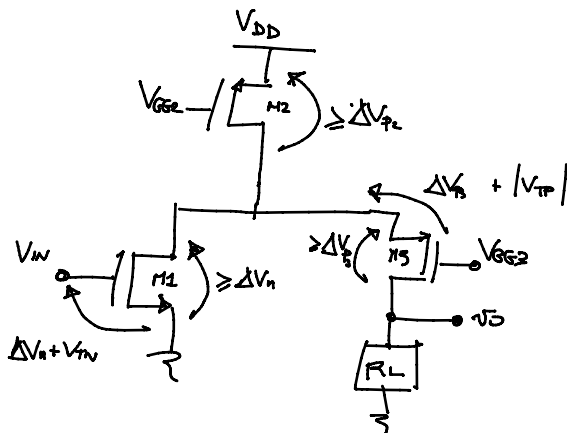
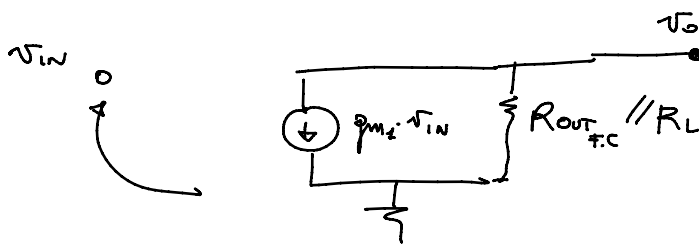
$$\underline{A_{V_{FCASC}}} = -g_{m1} \left(\underline{R_{o_{FC}}} \parallel R_L \right)$$

$$\underline{R_{o_{CG_{CASC}}} = \frac{G_{TCG}}{g_{o1} \cdot g_{o2}^{(CG)}}$$

$$; R_{o_{CG_{FCASC}}} = \frac{G_{TCG}}{(g_{o1} + g_{o2}) \cdot g_{o3}^{(CG)}}$$

$$R_{o_{FC}} < R_{o_{CASC}}$$

Folded Cascode



$$V_{GS} - V_{TN} = \Delta V_n \quad \text{NMOS}$$

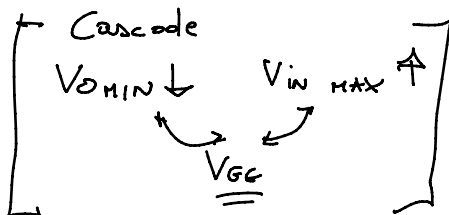
$$V_{SD} - |V_{TP}| = \Delta V_p \quad \text{PMOS}$$

$$V_{DD} \geq \Delta V_n + \Delta V_{p2}$$

$$\Delta V_n = V_{in} - V_{TN}$$

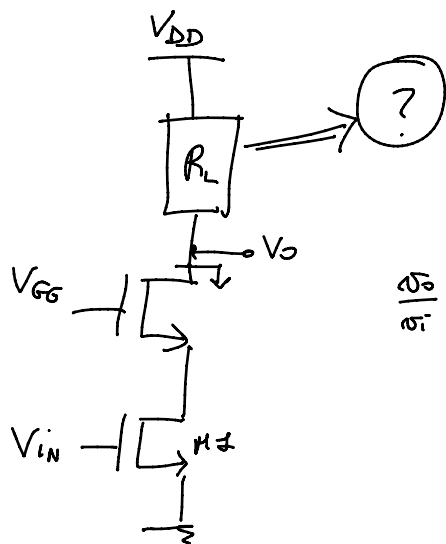
$$V_{DD} - \Delta V_p + V_{TN} \geq V_{in}$$

$$V_{in} \leq V_{DD} - \Delta V_{p2} + V_{TN} \triangleq V_{in \max} \uparrow$$

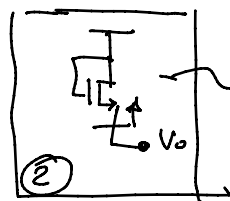
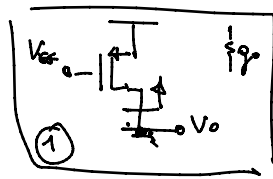
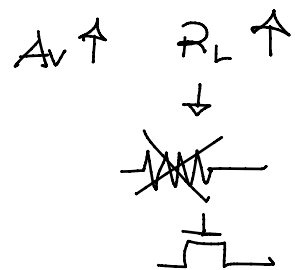


$$V_{DD} - V_O \geq \Delta V_{p2} + \Delta V_{p3}$$

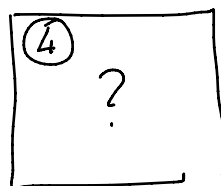
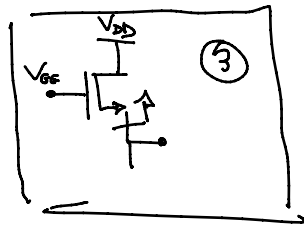
$$V_O \leq V_{DD} - \Delta V_{p2} - \Delta V_{p3} \triangleq V_{out \max} \uparrow$$



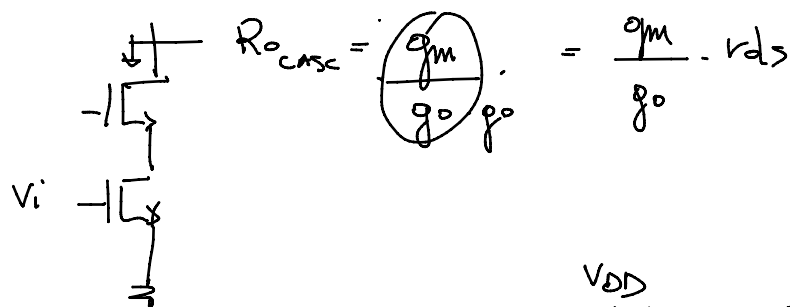
$$\frac{v_o}{v_i} = -g_{m2} (R_{o_{casc}} \parallel R_L)$$



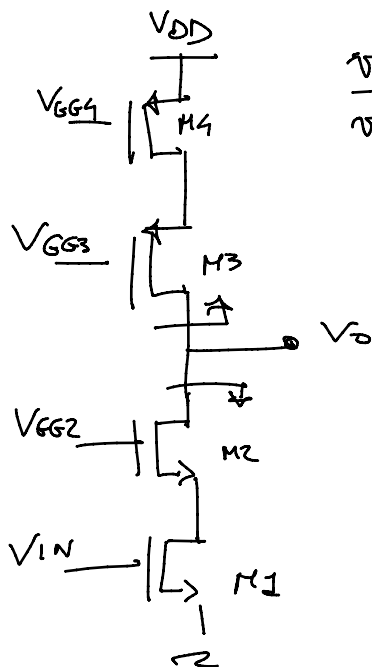
$\frac{1}{g_m} \Rightarrow R_L \downarrow$



100



$$R_{o_{casc}} = \frac{g_m}{g_o} = \frac{g_m}{g_o} \cdot v_{ds}$$



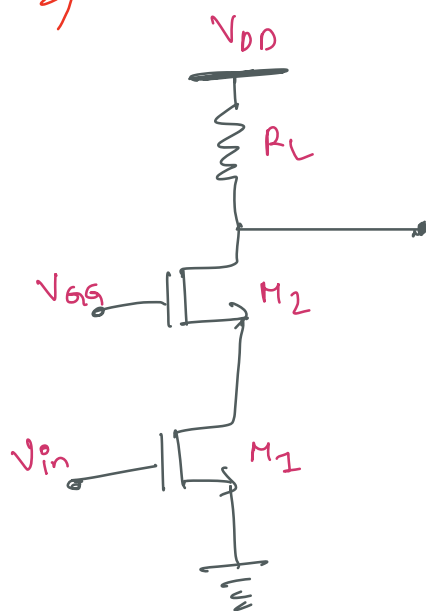
$$\frac{v_o}{v_{in}} = -g_{m1} (R_{o_{casc_N}} \parallel R_{o_{casc_P}})$$

10,000

→ Today we are going to continue the analysis of "Cascode configuration".

In the previous lecture we analyzed the cascode state with load equal to a pmos with constant gate voltage, whose equivalent resistance is equal to ($G_L = g_{o3}$) as shown in the above nodes.

Q₂) What if when the load is Arbitrary load?



R_L : Arbitrary load

Suppose if R_L is large

i.e $G_L = \frac{1}{R_L} \lll g_{o3}$

→ Let's start the analysis by calculating the "o/p Resistance" generated by the "Cascode Configuration" which will be in parallel to loading resistance (R_L) in the o/p Node

$\therefore R_{oc} = \text{o/p resistance of the cascode configuration}$

$$R_{oc} \Big|_{\substack{V_{in}=0 \\ R_L=\infty}} = \frac{V_o}{i_o} \Big|_{\substack{V_{in}=0 \\ R_L=\infty}} \Rightarrow R_{ocg} \Big|_{\substack{V_{G1}=0 \\ R_L=\infty}}$$

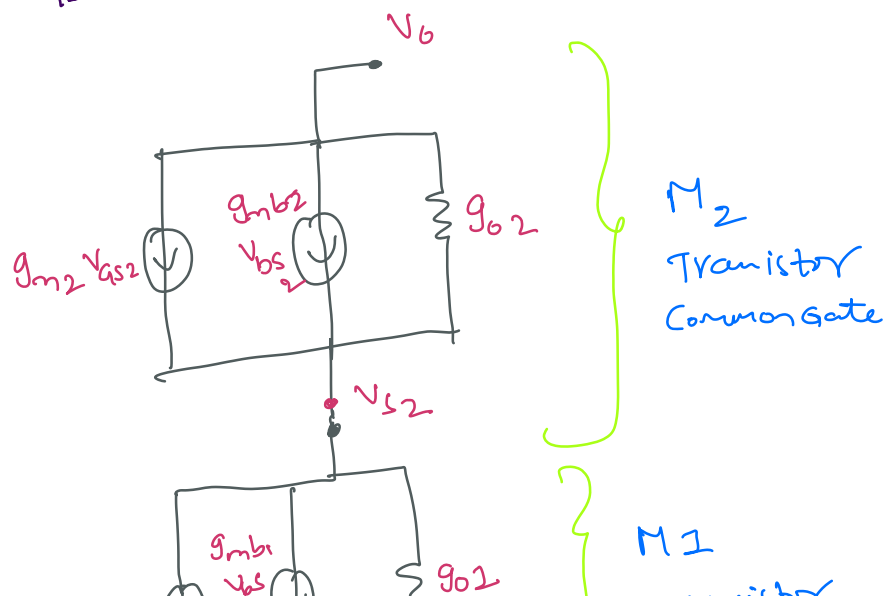
we were just focusing on
cascode stage

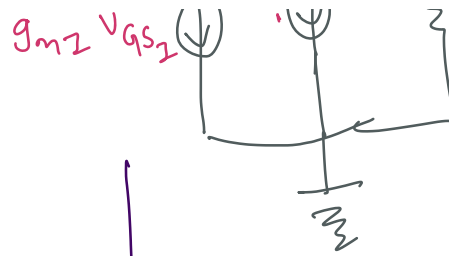
$\Rightarrow$

In other words what we
have to do is to calculate the o/p
Resistance of Common Gate configuration
with "i/p Gate voltage of $M_1 = 0$ "

$$\therefore R_{ocg} \Big|_{V_{G1}=0} =$$

Common
Gate SSA.





transistor
Common source

For M2
 $V_G = 0$ for small signal
 $V_S = V_{S2}$

$\therefore$ $V_{GS2} = V_{S2}$
 $V_{BS2} = V_{S2}$

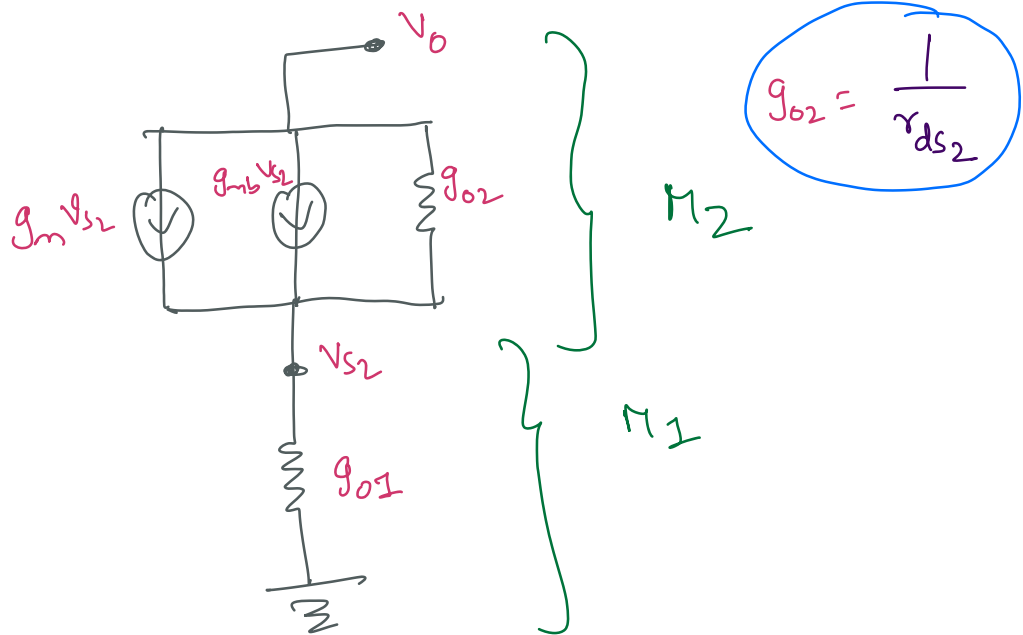
$V_G = V_B = 0$ Gated Bulk are
 Shorted $\therefore$

For M1, $V_{GS} = 0$
 because
 $V_G = V_{in} = 0$
 $V_S = 0$ (grounded)

$\& V_{BS} = 0$

$\therefore$ we will have only pure intrinsic
 op resistance of M1 i.e. g_{o1}

$\therefore$ The above circuit can be redrawn as



$g_{o2} = \frac{1}{r_{ds2}}$

$\rightarrow$ when we have studied the CG configuration
 we found that op Resistance is equal to

$R_{O_{CG}} = r_{ds2} \{ 1 + G_{M2} (r_{ds1}) \}$ "Source
Resistance
= Intrinsic"

(M₂)

Resistance
of (M₁)

$$G_{T2} \cdot r_{ds1} = (g_{m2} + g_{mb2} + g_{o2}) \frac{1}{g_{o1}}$$

$$\therefore g_m \gg g_o$$

we can say that the product of

$$G_{T2} \cdot r_{ds1} \gg 1$$

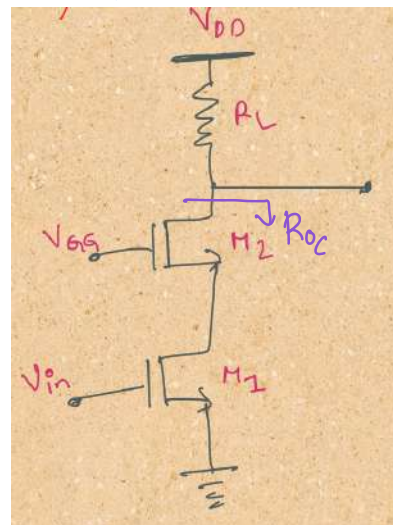
$$G_{T2} \approx g_{m2}$$

$$\therefore R_{o_{c_n}} = R_{oc} \approx \underbrace{g_{m2} \cdot r_{ds2} \cdot r_{ds1}}_{\text{approx. value}}$$

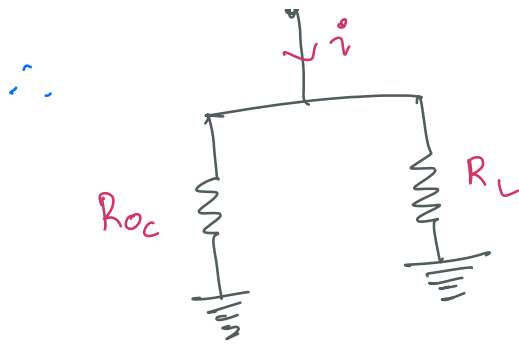
(M₂) (cathode
o/p
Resistance)

→ and now coming to the circuit the resistance we see at o/p node is parallel b/w

$R_{oc} \& R_L$ for ssc.



V_o



$\therefore$ Conductance seen from o/p node

$$G_{||} = \frac{1}{\underbrace{R_{||}}_{\text{parallel}}} = \frac{1}{R_{OC}} + \frac{1}{R_L}$$

$$\Rightarrow G_O = \frac{1}{g_{m2} \cdot r_{ds2} \cdot r_{ds1}} + G_L = \frac{g_{o1} \cdot g_{o2} + G_L}{g_{m2}}$$

(conductance seen from o/p node)

$r_{ds2} = g_{o2}$
 $r_{ds1} = g_{o1}$

$$\therefore G = \frac{g_{o1} \cdot g_{o2} + G_L \cdot g_{m2}}{g_{m2}}$$

$\rightarrow$ Now, let's compute the actual gain of this

Cascode Configuration.

$$A_{V_{\text{casc } R_L}} = \underbrace{A_{V1}_{CS}}_{\text{gain of CS}} \cdot \underbrace{A_{V2}_{CG}}_{\text{gain of CG}}$$

$$\text{gain of cascode stage with load } R_L = \frac{-g_{m1}}{g_{o1} + G_{L1}} \cdot \frac{G_{T2}}{G_L + g_{o2}}$$

A_{V2}_{CG}

G_{L1} is the load on common source stage i.e. The i/p Resistance of Common Gate stage. Because CS is loaded by CG.

$$\therefore G_{L1} = G_{i_{CG}} = \frac{G_{T2} \cdot G_L}{g_{o2} + G_L}$$

$$\therefore A_{V_{\text{casc } R_L}} = \frac{-g_{m1}}{g_{o1} + \frac{G_{T2} \cdot G_L}{g_{o2} + G_L}} \cdot \frac{G_{T2}}{G_L + g_{o2}}$$

$$A_{V_{casc}}^{R_L} = \frac{-g_{m1} \cdot G_{T2}}{g_{o1}(g_{o2} + G_L) + G_{T2} \cdot G_L}$$

* In the last class we calculated

$$A_{V_{casc}} \Big|_{R_L = r_{ds3}} = \frac{-g_{m1}}{g_{o3}}$$

⇒ let's verify the above equation with what we have derived today

we got

$$A_{V_{casc}}^{R_L} = \frac{-g_{m1} \cdot G_{T2}}{g_{o1}(g_{o2} + G_L) + G_{T2} \cdot G_L}$$

↓ Approx.

$$G_{T2} \approx g_{m2}$$

⇒

Earlier we assumed

$$G_L \ll g_{o3}, g_{o2}, g_{o1}$$

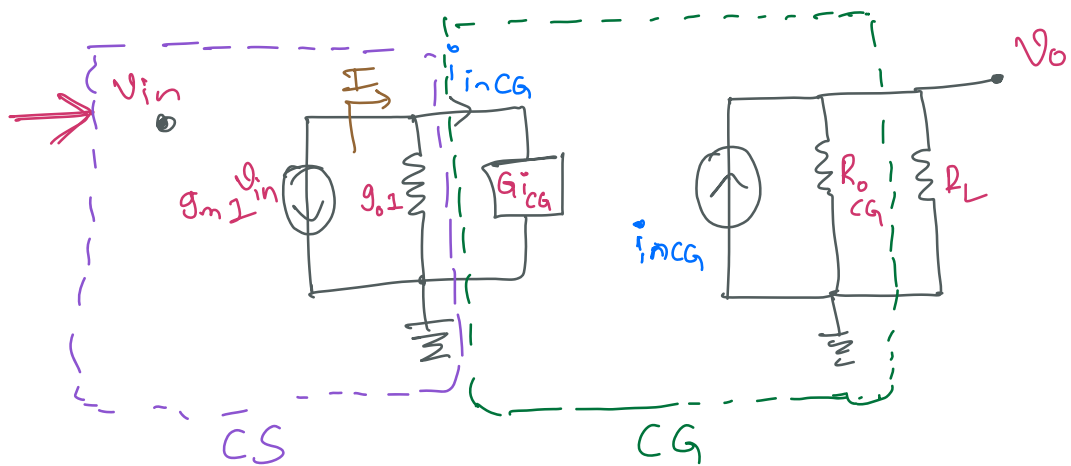
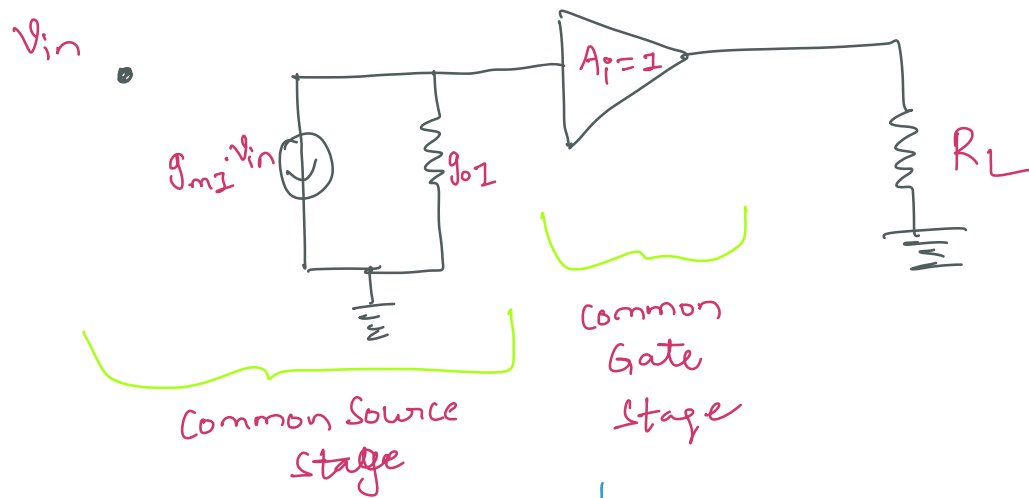
$$\frac{-g_{m1} \cdot g_{m2}}{g_{o1}(g_{o2}) + g_{m2} \cdot G_L}$$

↓ This term is $1/g''$ which we ...

calculating above

$$A_{v_{cascode R}} \Rightarrow -g_{m1} \cdot \frac{1}{G''} \Rightarrow -g_{m1} \cdot R''$$

* The cascode stage can be seen in the following way.



In CS (V_{in}) will generate current (I)

and current I can now travel into two paths

$$g_{o1} \text{ \& } G_{icG}$$

but as $G_{icG} \gg g_{o1}$ The total current is transferred to the CG stage

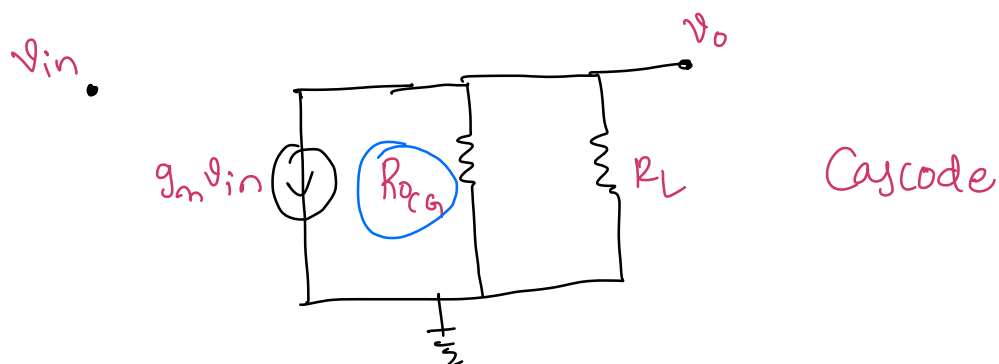
$$\therefore i_{inCG} \approx -g_{m1} \cdot v_{in}$$

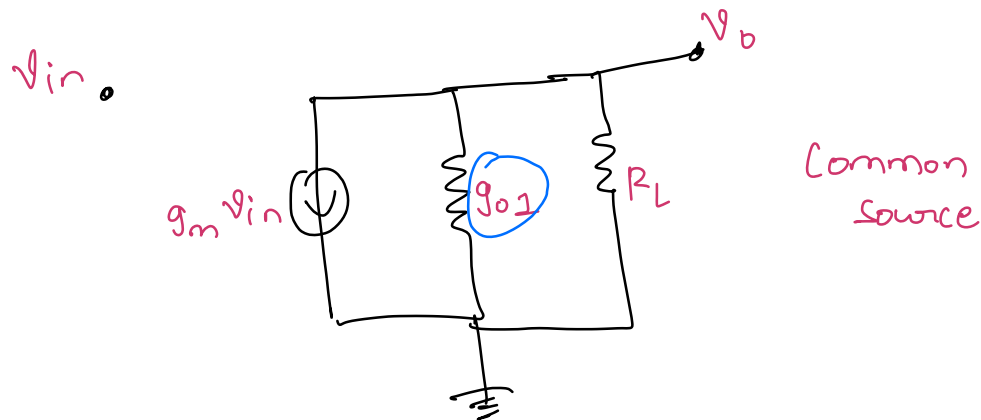
$$\therefore v_o = -g_{m1} \cdot v_{in} R''$$

This is the intuitive analysis of the Cascode stage.

→ we can see that with large R'' we can achieve "very large voltage gain".

→ The above representation of cascode stage can be further simplified.





The only difference we have in Cascode & CS stage is The R_{ocg} & $\frac{1}{g_{o1}}$

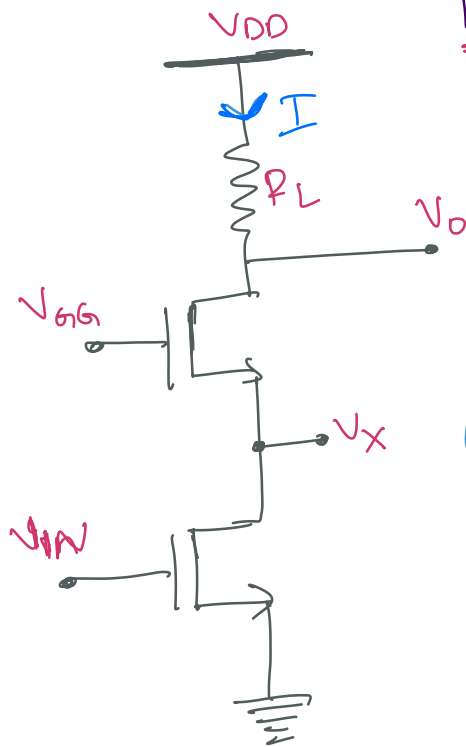
The "o/p Resistance in Cascode stage" is much more than in CS stage which is really beneficial.

→ But in Analog circuit design we should not forget all the small signal Analysis and conclusions we have done will depend on the operating point of the circuit.

therefore we should look at the ratio $\frac{g_m}{g_o} \uparrow 1$ which is true in SATURATION Region.

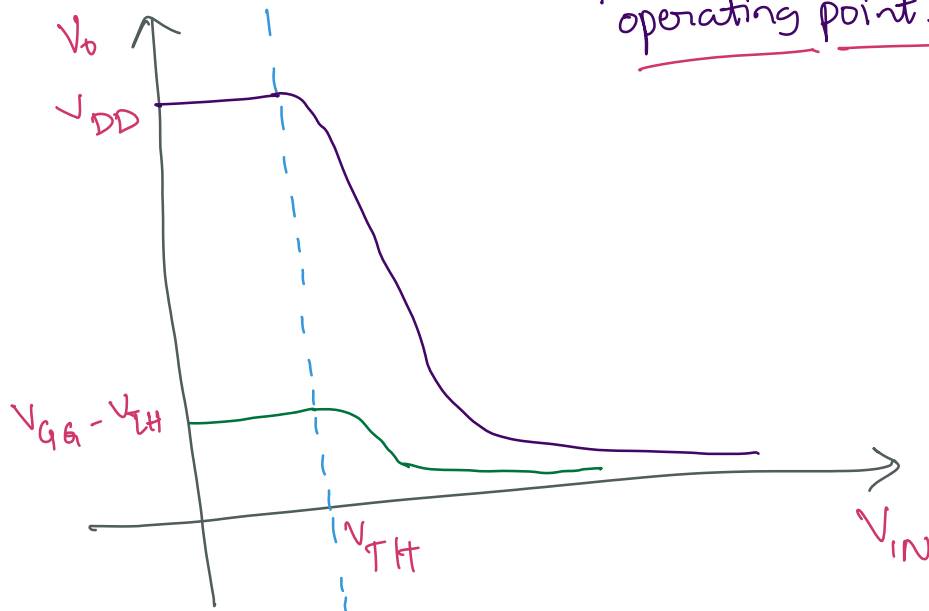
→ so we need to verify whether the circuit is

in saturation Region. So let's go for
Large signal Analysis



(we can also take into account the voltage V_X)

DC Analysis for operating point Analysis



(i) when input voltage for CS stage is

less than V_{IN} no current will flow through the circuit.

Then $V_0 = V_{DD}$

(ii) when $V_{IN} > V_{TH}$ the (M_1) stage will turn ON and current will flow through the circuit and there will be voltage drop across Resistor and the output voltage (V_0) will decrease rapidly at start and tend to be gradual as V_{IN} increases.

(iii) Q What is the value of V_x when device (M_2) is OFF?

Basically the value of V_x when no current is flowing should be equal (or) less than threshold.

$$V_G = V_{GG} \quad V_D = V_x \quad \text{for } M_2$$

$V_{GS} = V_{GG} - V_x \geq V_{th}$ for device to turn ON.

$$\Rightarrow V_{GG} - V_{th} \geq V_x \quad \text{for device}$$

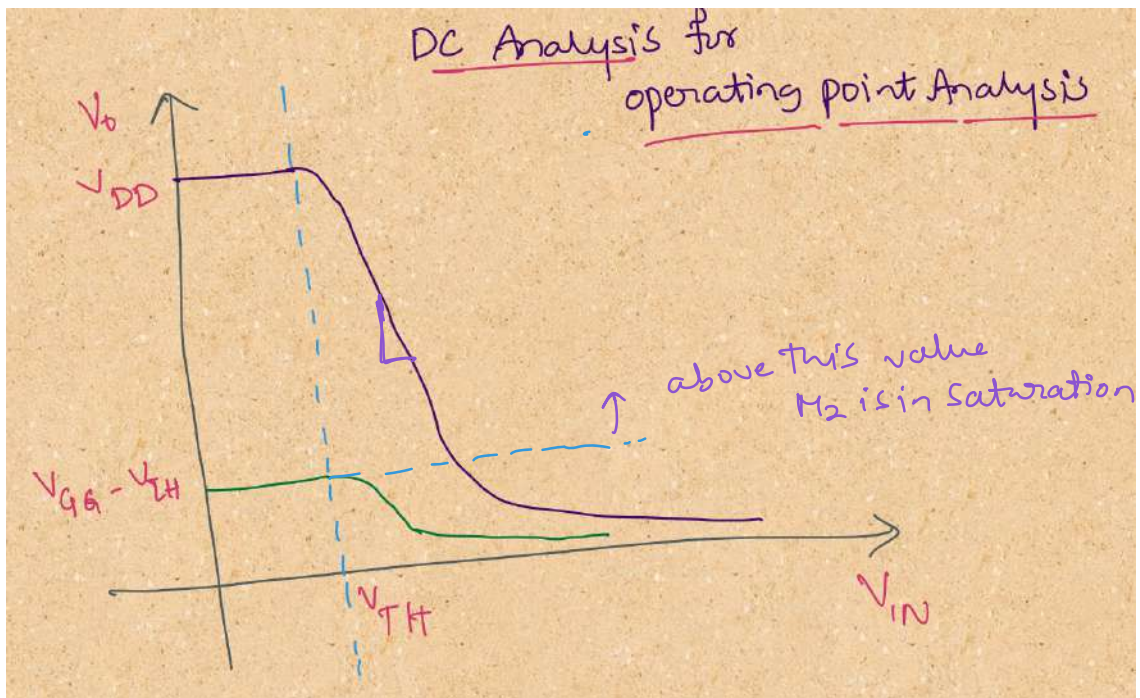
to turn on

$\therefore$ for M_2 to operate in saturation Mode
(i.e. to cope with current flow)

$$V_{GS} = V_{GG} - V_x \geq V_{th} \quad \text{and} \quad V_G = V_{GS}$$

is stuck at a constant value, $\therefore$

V_x should decrease.



$\rightarrow$ let's try to analyze saturation condition for M_1 & M_2 .

$$M_1 \text{ SAT} \Rightarrow V_x > \underbrace{V_{IN} - V_{TH1}}_{\text{overdrive of } M_1} \quad (V_{DS} \geq V_{GS} - V_{TH})$$

$$M_2 \text{ SAT} \Rightarrow V_D - V_X > \underbrace{V_{GG} - V_X - V_{TH2}}_{\substack{\text{overdrive of} \\ \Delta V_2 \text{ } M_2}}$$

↓

$$V_D > V_{GG} - V_{TH}$$

Q How do we set value of V_{GG} ?

Basically $V_{GG} - V_{TH}$ is the limit of saturation for M_2 .

So V_{GG} should be as low as possible.

~~But~~ But if V_{GG} is ^{so} small we may have some trouble with V_D .

Let's consider this aspect.

~~Till~~ Till now we got know that $V_D > V_{GG} - V_{TH}$
 This is the lower bound for V_D

Q How do we choose appropriate V_{GG} ?

For M_2 we need atleast OV of $M_2 + V_{TH2}$ for V_{GG}

M_1 we need atleast OV of M_2

OV: overdrive voltage.

Min

$$V_{0min} = \Delta V_1 + \Delta V_2$$

$$V_{0min} = \sqrt{\frac{2I_D}{\beta_1}} + \sqrt{\frac{2I_D}{\beta_2}}$$

$$V_{0min} \Rightarrow \sqrt{2I_D} \left[\frac{1}{\sqrt{\beta_1}} + \frac{1}{\sqrt{\beta_2}} \right]$$

↓ Minimum
up to this point the device will
be in saturation and gives us a Reasonable
Gain.

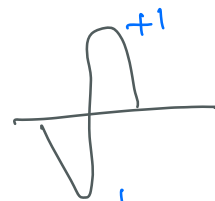
Actually it's not easy to full fill this
condition.

ex. let $A_V = -50V$

Range of input swing.
 $V_i = \pm 20mV$ (variations in the small signal)
↓
corresponding V_o

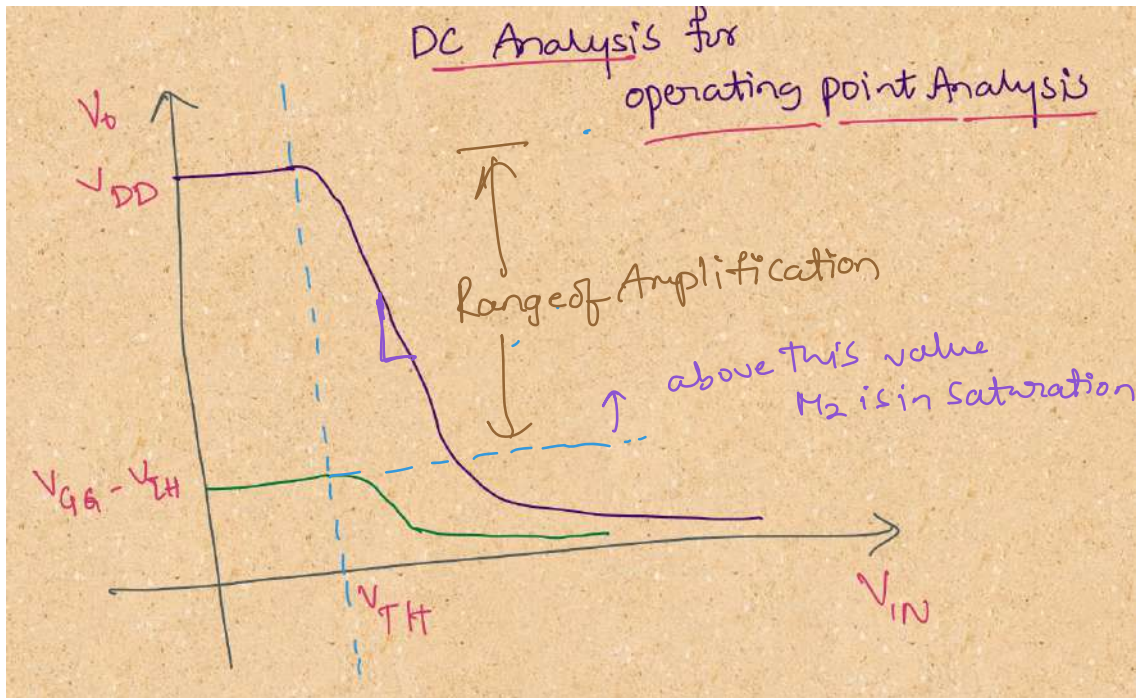
Range of op swing

$$V_o = \pm 1V \quad \text{i.e. } (2V)$$



which is the complete swing should

be compatible with the Range of Saturation Region.



Q what is the Range of i/p Swing That is acceptable. ?

$$V_O > V_{GG} - V_{T2} \triangleq V_{O_{min}}$$

Saturation of M_1 is tightly related to applied input V_{in}

$$\boxed{V_{in} = \Delta V_1 + V_{T1}}$$

(should be atleast)

→ we have

$$V_{GG} \geq \Delta V_2 + V_{T2} + \Delta V_1$$

$$V_{GG} \geq \Delta V_2 + V_{T2} + V_{IN} - V_{T1}$$

$$V_{IN} \leq V_{GG} - \Delta V_2 + \underbrace{V_{T2} - V_{T1}}_{\text{almost zero}}$$

$$\Rightarrow \boxed{V_{IN} \leq V_{GG} - \Delta V_2} \triangleq \boxed{V_{inMax}}$$

This is the Maximum value
for V_{IN}

which restricts the value of
i/p Swing.

(i) output voltage swing & Input swing Restriction

$$V_o \geq V_{GG} - V_{T2} \triangleq V_{o\min}$$

$$V_{in} \leq V_{GG} - \Delta V_2 \triangleq V_{in\max}$$

for $V_o \rightarrow V_{GG}$ should be small
so that saturation region is large

$V_{in} \rightarrow V_{GG}$ should be high
so that i/p swing can be large

There are contradicting requirements

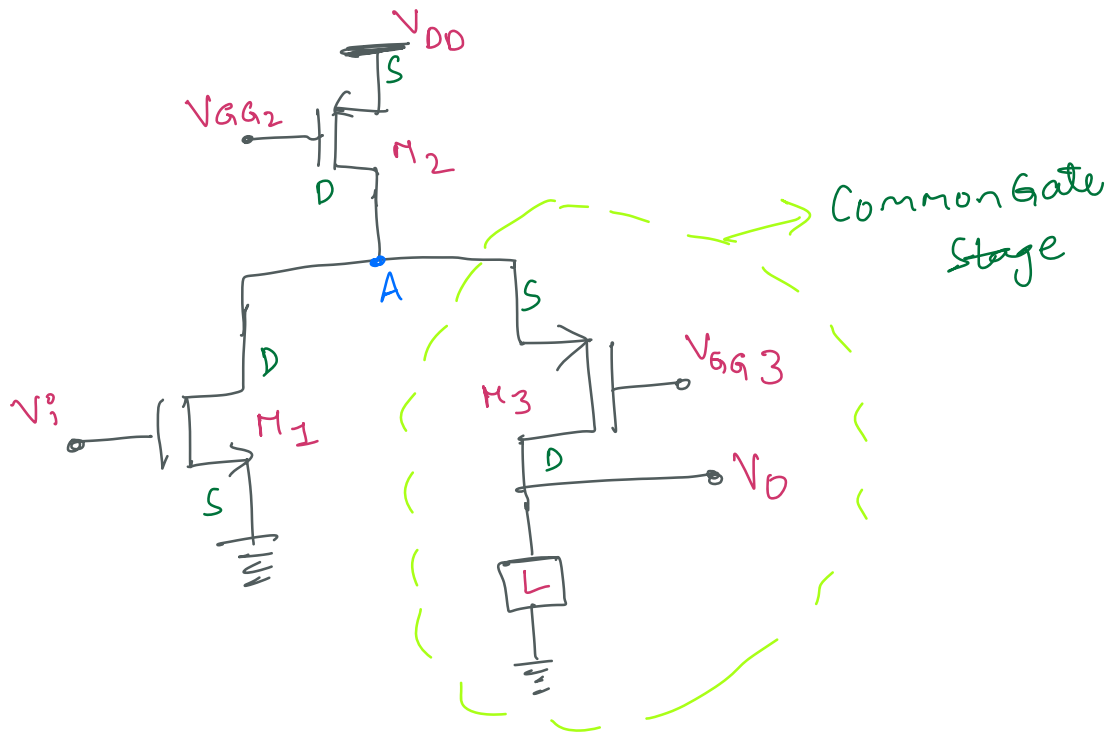
so we should find a Balance

b/w V_o & V_{in} by choosing

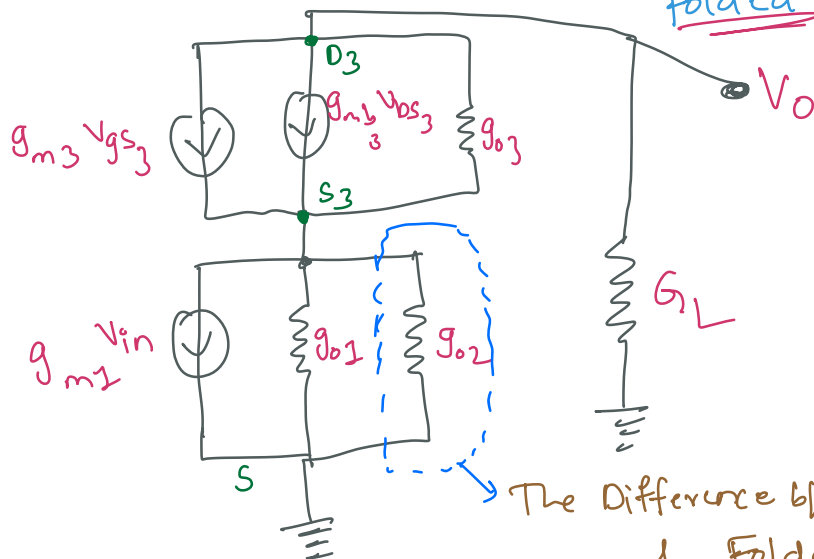
an optimal V_{GG}

This is a quite Annoying Problem.

$\rightarrow$ To prevent this issue in cascode configuration,
we go for "Folded Cascode Configuration"



* As usual let's start with SSA of the Folded configuration.



SSM of NMOS & PMOS are same.

The Difference b/w Normal Cascode & Folded cascode

For M1

$$\begin{aligned} V_{gs} &= V_g - V_s \\ &= V_{in} - 0 \\ &= V_{in} \end{aligned}$$

For M2

$V_g = V_{GG}, V_s = V_{DD}$
But in small signal world both constant voltages are

grounded.

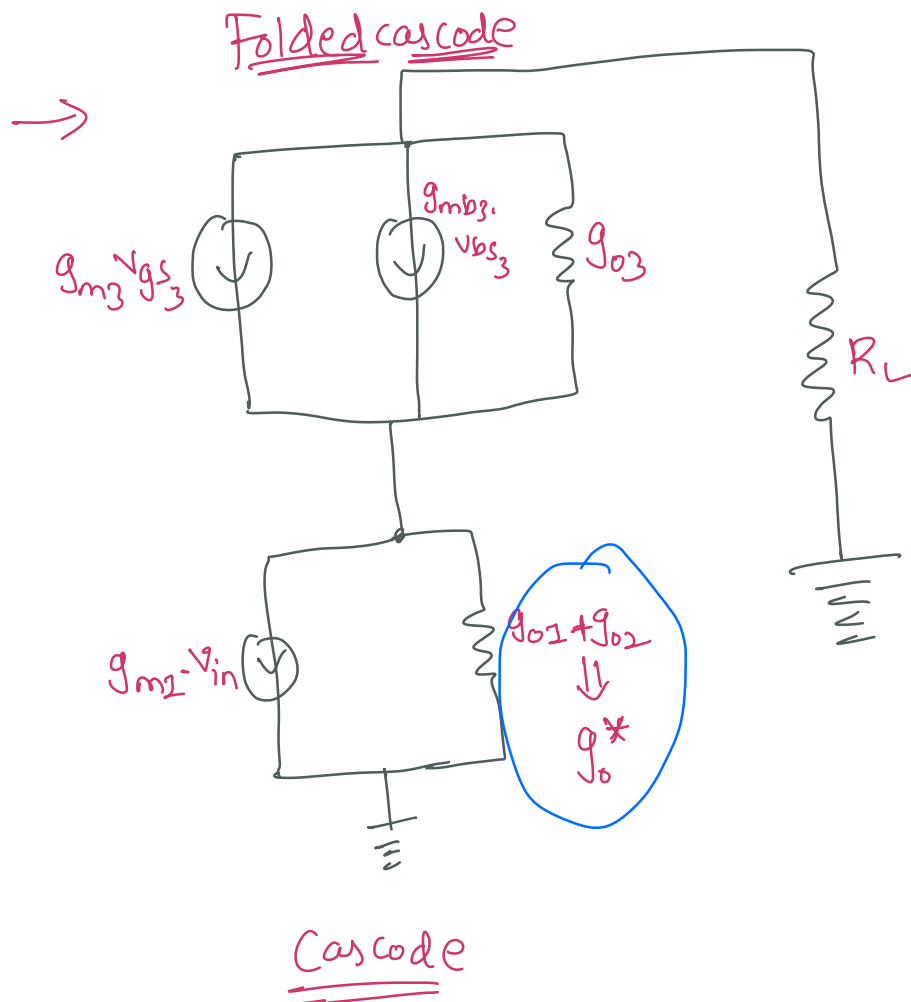
$$\therefore V_{gs} = 0 - 0 = 0$$

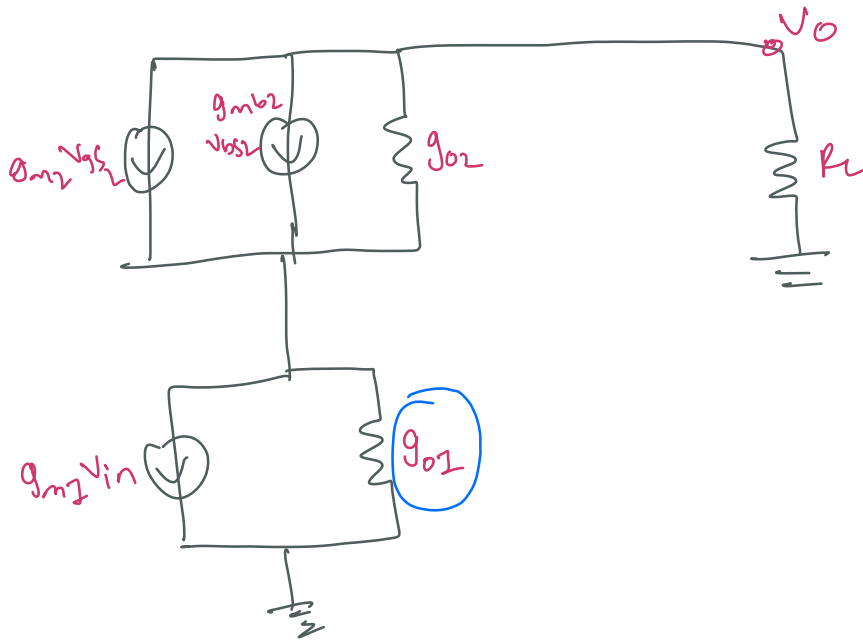
$$V_{bs} = 0$$

M_2 only has the intrinsic o/p conductance (g_{o2}) of when we look from Node A and it is in parallel with g_{o1} because it is connected from Node A to Ground.

For M_3

Both PMOS & NMOS have similar small signal Model.





→ The only difference b/w Normal and Folded cascode is that we have double g_o in Folded cascode

For Folded cascode stage

$$A_{V1} = \frac{-g_{m1}}{g_{o1} + g_{o2} + G_{i_{CG}}}$$

FC
(for M_1)

$G_{i_{CG}}$: i/p Conductance of CG stage

$$A_V = -g_{m1} (R_{oCG} \parallel R_L)$$

FCAS
for Folded
cascode
stage

↓
This what varies in Folded cascode stage

Incascode

$$A_{V1} = \frac{-g_{m1}}{g_{o1} + g_{i_{CG}}}$$

$$A_{V_{casc}} = -g_{m1} (R_{o_{CG}} \parallel R_L)$$

$$R_{o_{CG}} = \frac{G_{TCG}}{g_{o1} \cdot \underbrace{g_{o2}}_{\text{CG stage}}}$$

Folded cascode

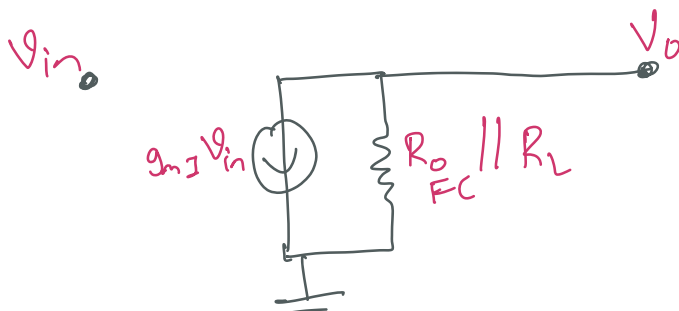
$$A_{V1} = \frac{-g_{m1}}{g_{o1} + g_{o2} + g_{i_{CG}}}$$

$$A_{V_{FCAS}} = -g_{m1} \cdot (R_{o_{CG, FCAS}} \parallel R_L)$$

output resistance of FC

$$R_{o_{CG}} = \frac{G_{TCG}}{(g_{o1} + g_{o2}) \underbrace{(g_{o2})}_{\text{CG stage}}}$$

→ As far as SSN of Folded Cascode is concerned.



Q

Q

Which is the larger gain among
Cascode & Folded Cascode?

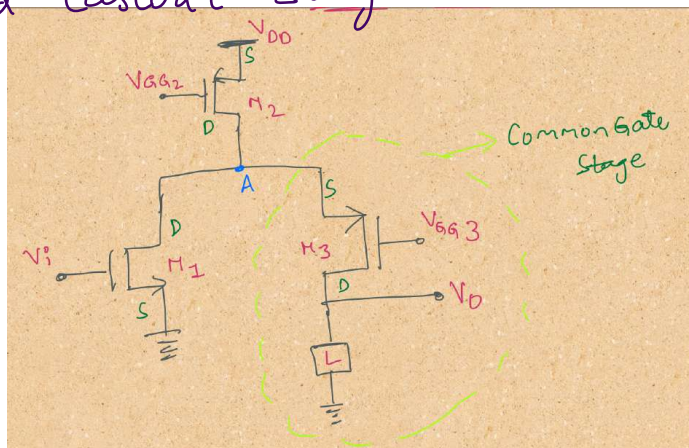
If we observe the o/p Resistance of
FC is less than Cascode stage.

$\therefore$ The Gain of FC is less than
Cascode stage.

Q

Why do we use FC instead of Cascode?
The answer is in Large signal Model.

Let's study the i/p and o/p swing of Folded
cascode Amplifier to see the benefits of
Folded cascode stage.



$V_{GG2} + V_{GG3}$
are constant
values

→ large signal Model of Folded cascode stage.

$$V_{GS} - V_{TN} = \Delta V_n \quad \text{for NMOS}$$

$$V_{SG} - |V_{TP}| = \Delta V_p \quad \text{for PMOS}$$

Saturation for NMOS

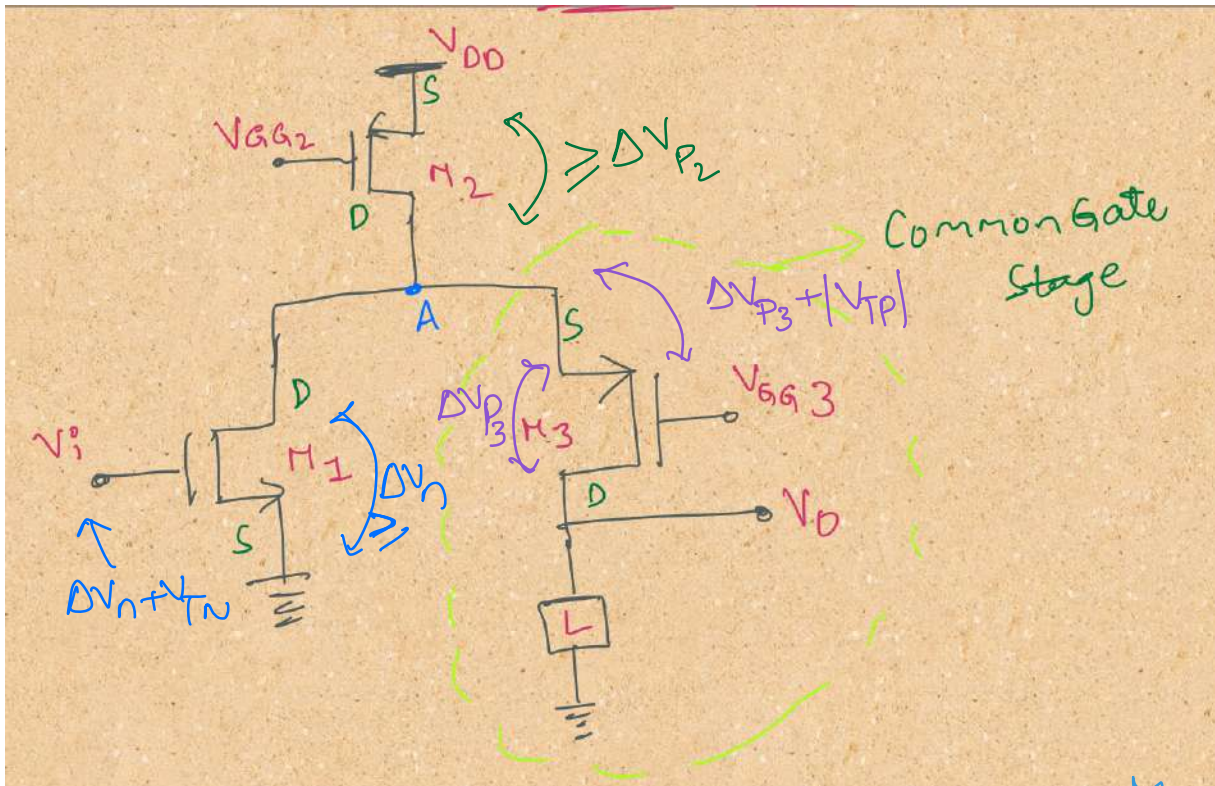
$$V_{DS} \geq V_{GS} - V_{TH}$$

$$\therefore V_{OV} \geq V_{in} - V_{TN}$$

$$V_{in} \leq V_{OV} + V_{TN}$$

$$\Rightarrow V_{in} \leq \Delta V_n + V_{TN}$$

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→ If we look at node A both M_1 & M_2

$$V_{DD} > \Delta V_n + \Delta V_{P2}$$

$$\Delta V_n = V_{in} - V_{TN}$$

$$\Rightarrow \boxed{V_{DD} - \Delta V_{P2} + V_{TN} \geq V_{in}} \equiv V_{in \max}$$

→ If we look at M_2 & M_3

$$V_{DD} - V_O \geq \Delta V_{P2} + \Delta V_{P3}$$

$$\Rightarrow V_O \leq V_{DD} - \Delta V_{P2} - \Delta V_{P3}$$

V_O
 (max)

In Folded cascode case

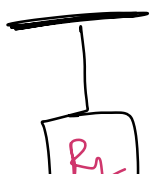
V_{Omax} & V_{imax} are not linked together.

Q What do you think is the best configuration?

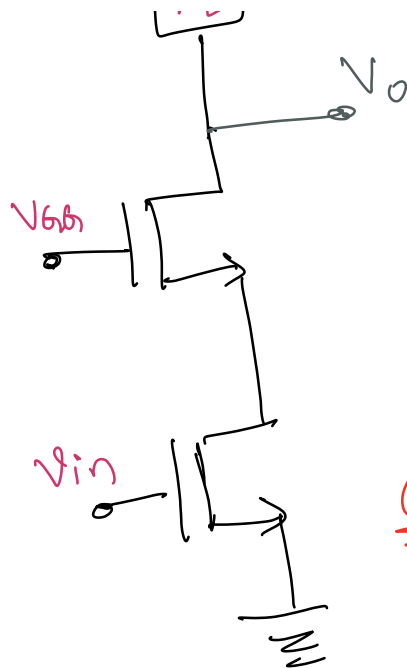
Ans As far as Gain is considered CASCODE is good.

As far as i/p & o/p swing is considered Folded cascode is used.

Quiz (Q) How can we replace R_L to achieve more gain?



we have



$$\frac{V_o}{V_{in}} = -g_{m2} (R_{o\text{cas}} \parallel R_L)$$

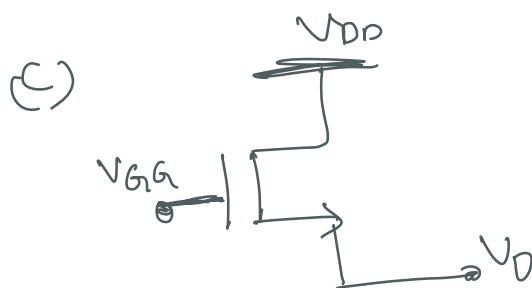
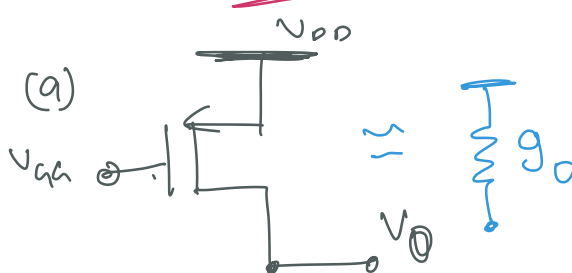
To increase Gain

$$A_v \uparrow$$

$$R_L \uparrow$$

Q What will you replace R_L with to achieve high Resistance?

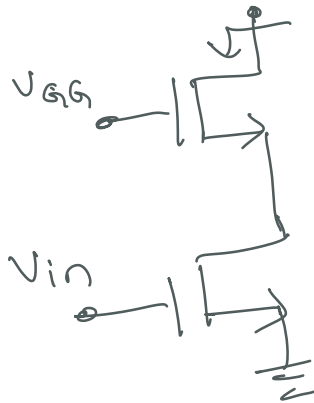
option



(d) cascode configuration reaches Highest o/p Resistance

we need to calculate "o/p Resistance" of all the above cases

(d) cascode stage o/p resistance



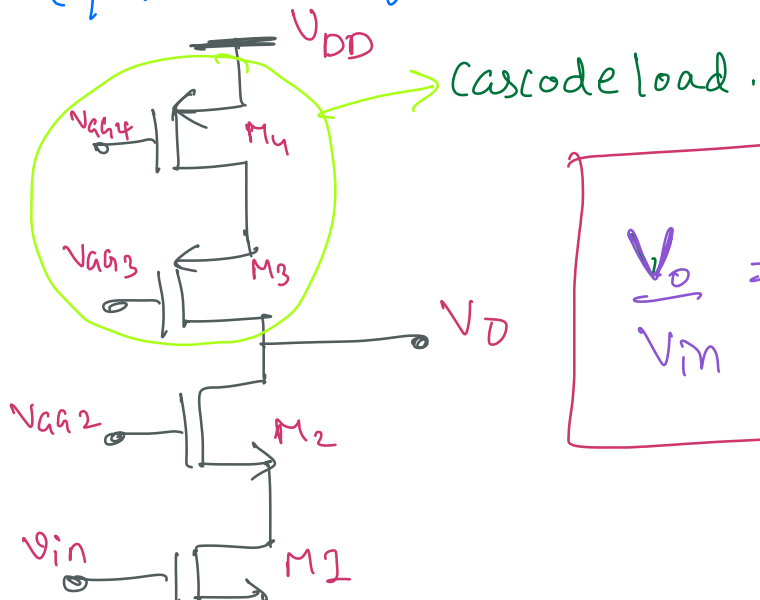
$$R_{o_{casc}} = \frac{g_m}{g_o \cdot g_o} = \left(\frac{g_m}{g_o} \right) r_{ds}$$

This is greater value

So we can use cascode stage as our load Resistance

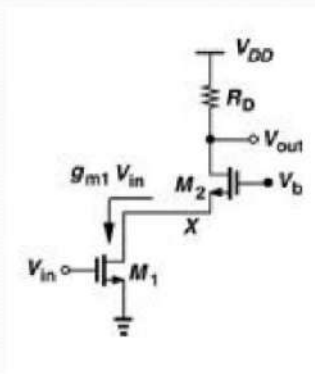
→ ~~Q~~ But cascode stage is what we use for Amplification and how do we use it for loading to.?

Basically we make use of PMOS equivalent of cascode as shown below.



$$\frac{V_o}{V_{in}} = -g_{m1} \cdot \left(R_{o_{casc N}} \parallel R_{o_{casc P}} \right)$$

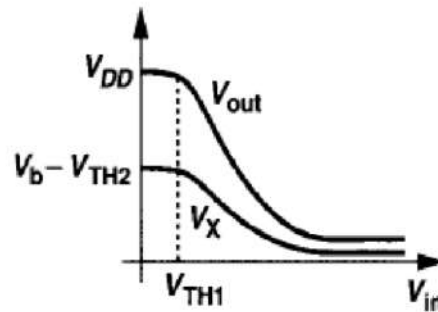
MOSFET Cascode



$$A_v \approx g_{m1} \{ [r_{o1} r_{o2} (g_{m2} + g_{mb2})] \parallel R_D \}$$

$$R_{out} = \{ [1 + (g_{m2} + g_{mb2}) r_{o2}] r_{o1} + r_{o2} \} \parallel R_D$$

$$\approx [r_{o1} r_{o2} (g_{m2} + g_{mb2})] \parallel R_D$$

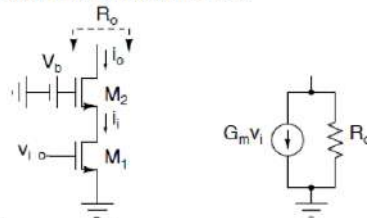


RZ

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Cascode Stage

- Combines a common-source stage and a common-gate stage to achieve both voltage gain of CS and wide bandwidth of CG.



$$G_m = \frac{i_o}{v_i} = g_{m1} \times \frac{i_o}{i_i} \approx g_{m1}, \quad R_o \approx r_{o2} (1 + g_{m2}' r_{o1})$$

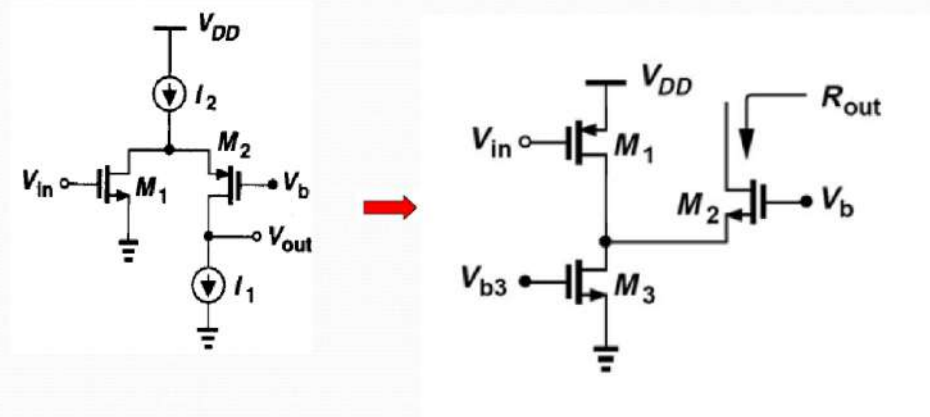
$$G_m R_o \approx g_{m1} r_{o2} (1 + g_{m2}' r_{o1}) \approx g_{m1} r_{o1} \times g_{m2}' r_{o2} \approx (g_m r_o)^2$$

$$\frac{v_x}{v_i} = -g_{m1} R_x \approx -\frac{g_{m1}}{g_{m2}'}$$

Low voltage gain for the first stage

28

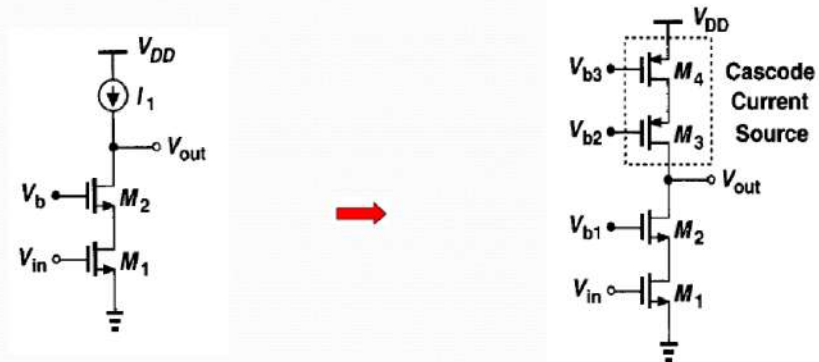
Folded Cascode



RZ

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Cascode with Cascode Load



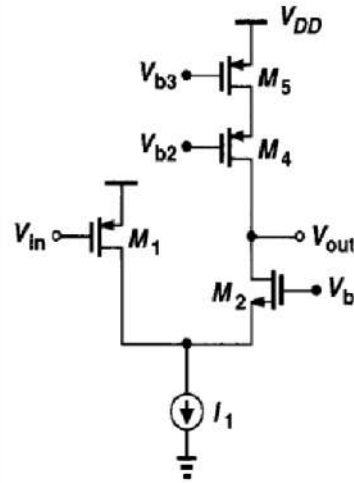
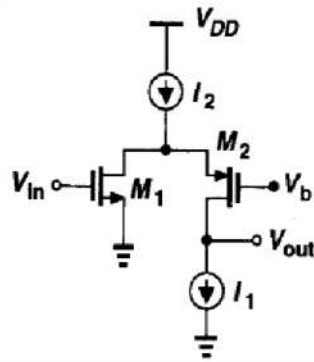
$$R_{out} = \{[1 + (g_{m2} + g_{mb2})r_{O2}]r_{O1} + r_{O2}\} || \{[1 + (g_{m3} + g_{mb3})r_{O3}]r_{O4} + r_{O3}\}$$

$$A_v \approx g_{m1}[(g_{m2}r_{O2}r_{O1}) || (g_{m3}r_{O3}r_{O4})]$$

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Folded Cascode with Cascode Load



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